

CE2205 – Mechanics of Materials
Spring 2026
Final Project:
/100pts

Due date: 4/30 at 9.00am

GANTT Chart – DUE Friday 3/20 at 12.00pm

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Introduction

This project analyzes the structural behavior of a single frame within a storage shed system using principles from statics and mechanics of materials. The frame consists of a steel beam supported by wood columns and is subjected to multiple loading conditions, including dead, live, wind, and snow loads. Using ASCE load combinations, the governing load cases were determined and converted from area loads to equivalent line loads acting on the beam. Shear and bending moment diagrams were developed and used for further stress analysis. Eventually, leading to a change in support system and using our own engineering judgment to make a structural recommendation for our clients.

The structure is a simple building with a flat roof, composed of several main frames, as shown in Figures. 1-3.

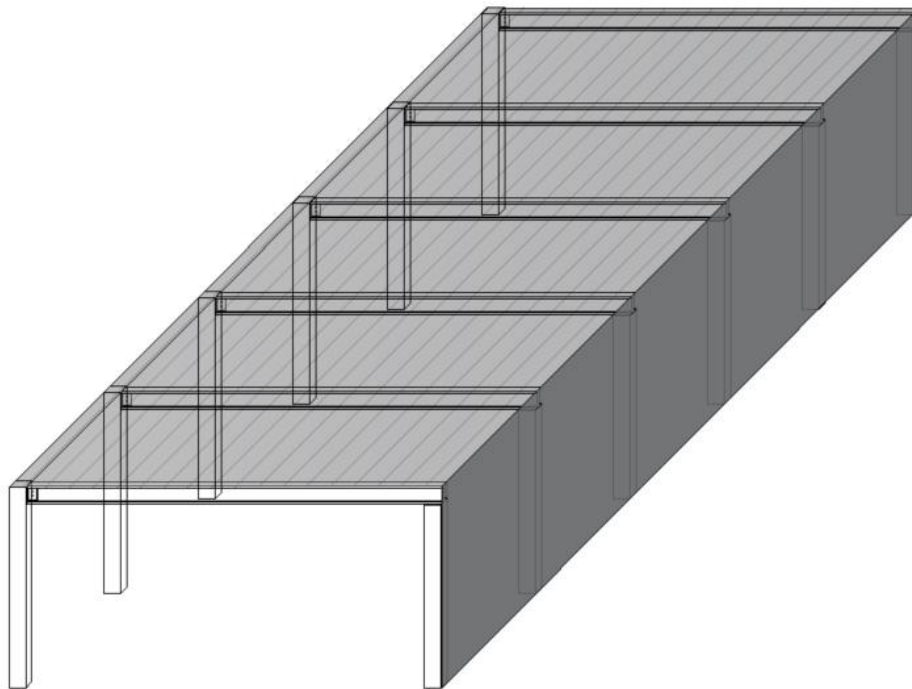


Figure 1: 3-D model of shed (Absi).

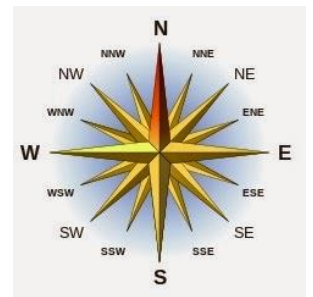
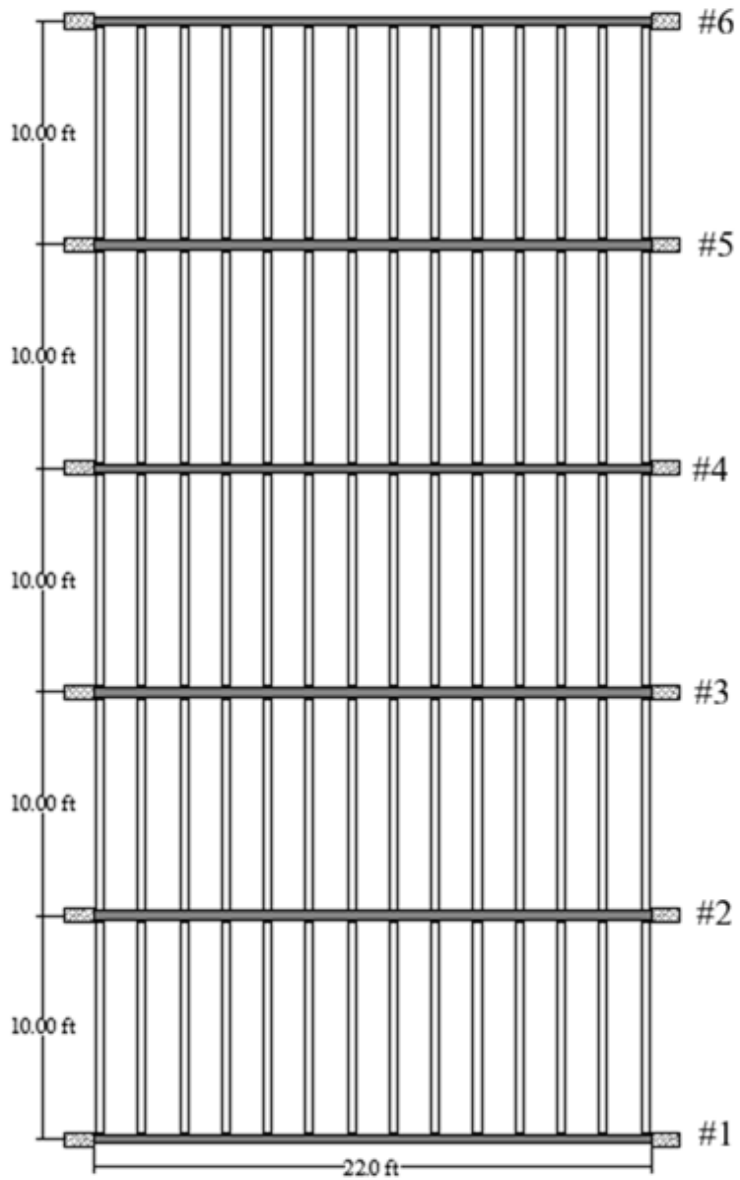


Figure 2: Top view of shed (Absi).

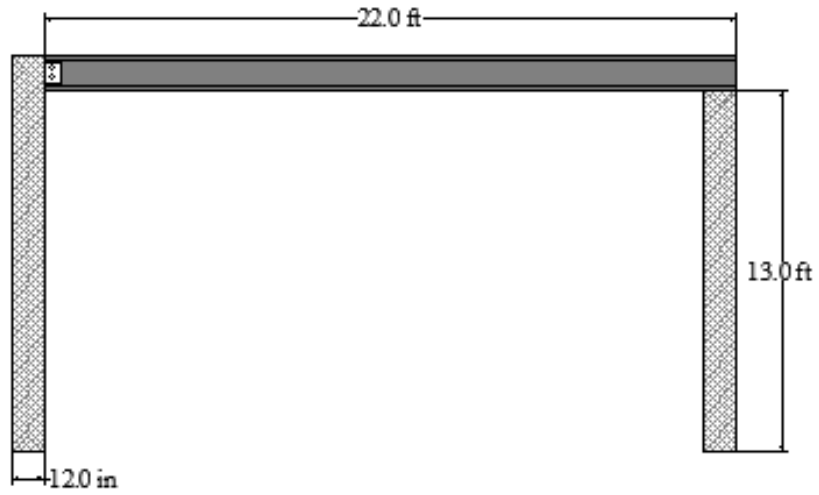


Figure 3: Elevation of Frame 2 (Absi).

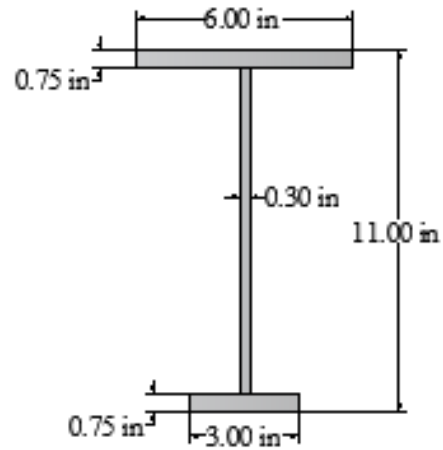


Figure 4: Beam Cross-Section (Absi).

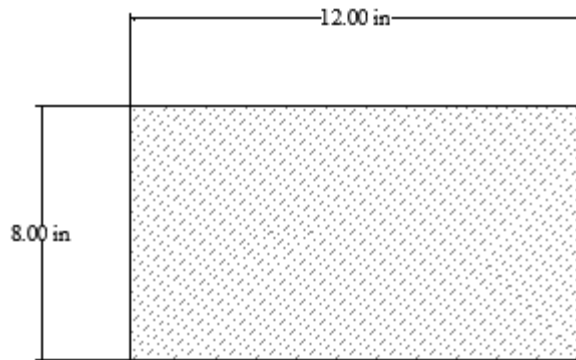


Figure 5: Column Cross-Section (Absi).

Step 1: GANTT Chart (5 pts) – DUE Friday 3/20 at 11.59pm

The GANTT chart allowed us as a group to stay on track and understand our own workloads. It additionally allowed us to establish and stick to clear future deadlines, enabling us to hold ourselves and each other accountable. Furthermore, the GANTT chart gave each of us a visual of where our partner was at with their work, enabling both of us to check in with the other if we noticed we were falling behind schedule, and further subdivide work if needed.

Step 2: Load Combinations Calculations (15 pts)

a) Dead Load

$$A_{beam} = A_{top\ flange} + A_{web} + A_{bottom\ flange}$$

$$A_{beam} = (0.75\ in \cdot 6in) + (0.3in \cdot 9.5in) + (0.75in \cdot 3in) = 9.6\ in^2$$

$$w_{beam} = \left(0.284\ \frac{lb}{in^3}\right)(9.6\ in^2) \approx 2.73\ \frac{lb}{in}$$

$$2.73\ \frac{lb}{in} \cdot \frac{12\ in}{ft} = 32.7\ \frac{lb}{ft}$$

b) Load Combinations

1. $1.4D$

$$1.4(232.7\ lb/ft)$$

2. $1.2D + 1.6L + 0.5S$

$$1.2(232.7\ lb/ft) + 1.6 \begin{cases} 150\ lb/ft, & 0 \leq x \leq 22 \\ 500\ lb \cdot ft, & x = 11 \end{cases} + 0.5 \begin{cases} 400 - \frac{300x}{14.667}\ lb/ft, & 0 \leq x \leq 14.667 \\ 100\ lb/ft, & 14.667 \leq x \leq 22 \end{cases}$$

3A. $1.2D + 1.6S + L$

$$1.2(232.7\ lb/ft) + 1.6 \begin{cases} 400 - \frac{300x}{14.667}\ lb/ft, & 0 \leq x \leq 14.667 \\ 100\ lb/ft, & 14.667 \leq x \leq 22 \end{cases} + \begin{cases} 150\ lb/ft, & 0 \leq x \leq 22 \\ 500\ lb \cdot ft, & x = 11 \end{cases}$$

3B. $1.2D + 1.6S + 0.8W_1$

$$1.2(232.7\ lb/ft) + 1.6 \begin{cases} 400 - \frac{300x}{14.667}\ lb/ft, & 0 \leq x \leq 14.667 \\ 100\ lb/ft, & 14.667 \leq x \leq 22 \end{cases} - 0.8(300\ lb/ft)$$

3C. $1.2D + 1.6S + 0.8W_2$

$$1.2(232.7\ lb/ft) + 1.6 \begin{cases} 400 - \frac{300x}{14.667}\ lb/ft, & 0 \leq x \leq 14.667 \\ 100\ lb/ft, & 14.667 \leq x \leq 22 \end{cases} + 0.8 \begin{cases} 350\ lb/ft, & 0 \leq x \leq 11 \\ 200\ lb/ft, & 11 \leq x \leq 22 \\ 1043.75\ lb, & x = 22 \end{cases}$$

4A. $1.2D + 1.6W_1 + L + 0.5S$

$$1.2(232.7\ lb/ft) - 1.6(300\ lb/ft) + \begin{cases} 150\ lb/ft, & 0 \leq x \leq 22 \\ 500\ lb \cdot ft, & x = 11 \end{cases} + 0.5 \begin{cases} 400 - \frac{300x}{14.667}\ lb/ft, & 0 \leq x \leq 14.667 \\ 100\ lb/ft, & 14.667 \leq x \leq 22 \end{cases}$$

4B. $1.2D + 1.6W_2 + L + 0.5S$

$$1.2(232.7\ lb/ft) + 1.6 \begin{cases} 350\ lb/ft, & 0 \leq x \leq 11 \\ 200\ lb/ft, & 11 \leq x \leq 22 \\ 1043.75\ lb, & x = 22 \end{cases} + \begin{cases} 150\ lb/ft, & 0 \leq x \leq 22 \\ 500\ lb \cdot ft, & x = 11 \end{cases} + 0.5 \begin{cases} 400 - \frac{300x}{14.667}\ lb/ft, & 0 \leq x \leq 14.667 \\ 100\ lb/ft, & 14.667 \leq x \leq 22 \end{cases}$$

$$5. 1.2D + L + 0.2S$$

$$1.2(232.7 \text{ lb/ft}) + \begin{cases} 150 \text{ lb/ft}, & 0 \leq x \leq 22 \\ 500 \text{ lb} \cdot \text{ft}, & x = 11 \end{cases} + 0.2 \begin{cases} 400 - \frac{300x}{14.667} \text{ lb/ft}, & 0 \leq x \leq 14.667 \\ 100 \text{ lb/ft}, & 14.667 \leq x \leq 22 \end{cases}$$

$$6A. 0.9D + 1.6W_1$$

$$0.9(232.7 \text{ lb/ft}) - 1.6(300 \text{ lb/ft})$$

$$6B. 0.9D + 1.6W_2$$

$$0.9(232.7 \text{ lb/ft}) + 1.6 \begin{cases} 350 \text{ lb/ft}, & 0 \leq x \leq 11 \\ 200 \text{ lb/ft}, & 11 \leq x \leq 22 \\ 1043.75 \text{ lb}, & x = 22 \end{cases}$$

$$7. 0.9D$$

$$0.9(232.7 \text{ lb/ft})$$

Full Work: [Load Combination Work](#)

c) Loading Diagrams

Graphs not drawn by hand generated in SkyCiv and enhanced by AI graphics:

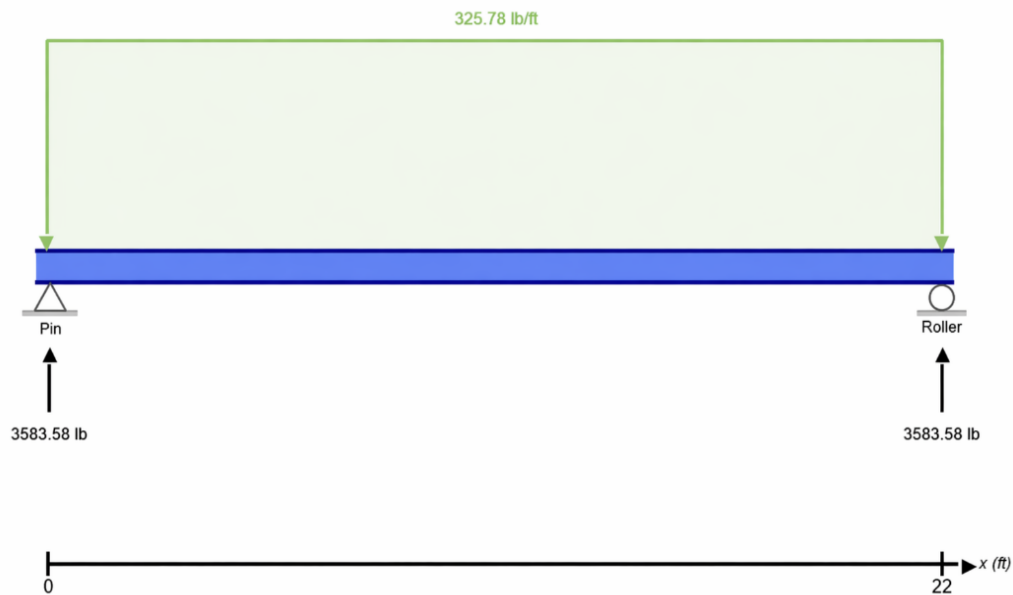


Figure 6 - Load Combination One: $1.4D$

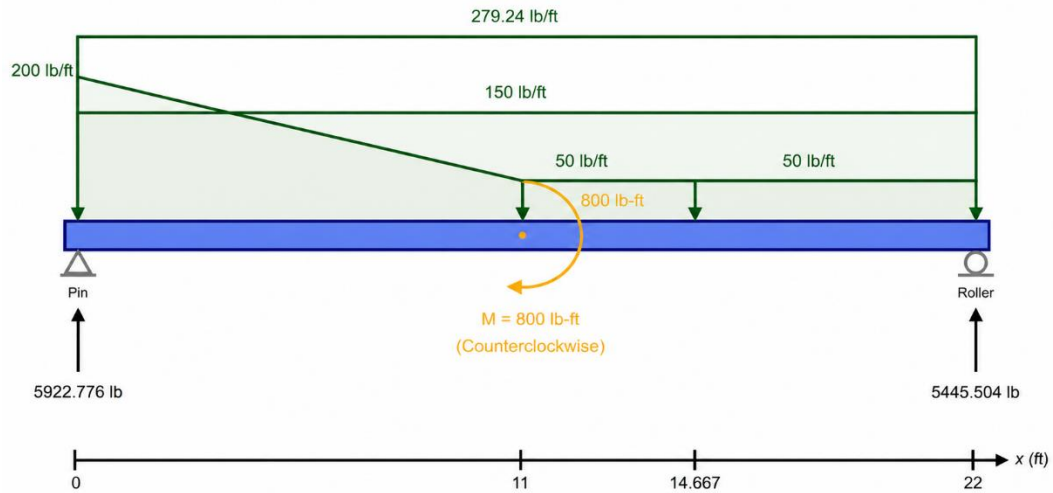


Figure 7 - Load Combination Two: $1.2D + 1.6L + 0.5S$

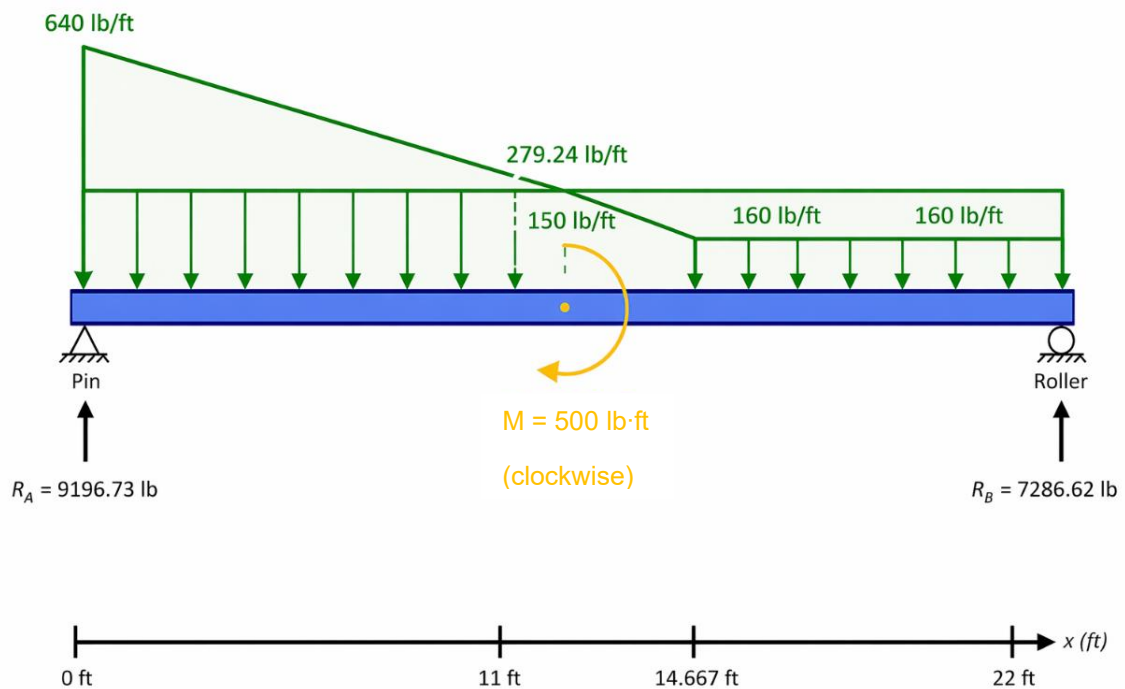


Figure 8 - Load Combination 3A: $1.2D + 1.6S + L$

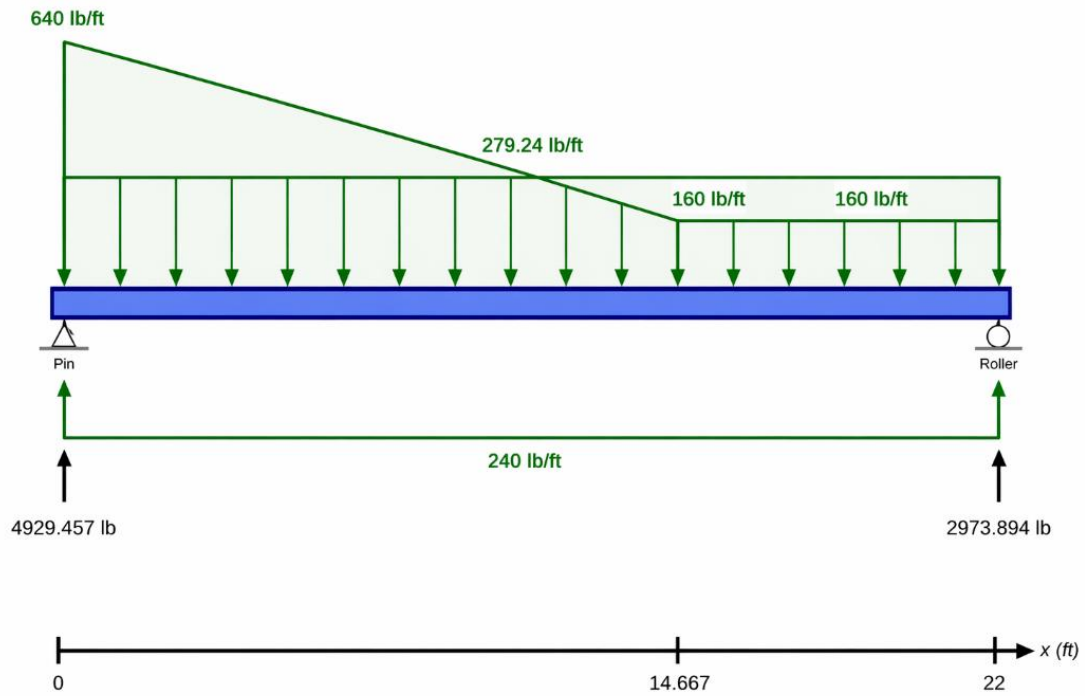


Figure 9 - Load Combination 3B: $1.2D + 1.6S + 0.8W_1$

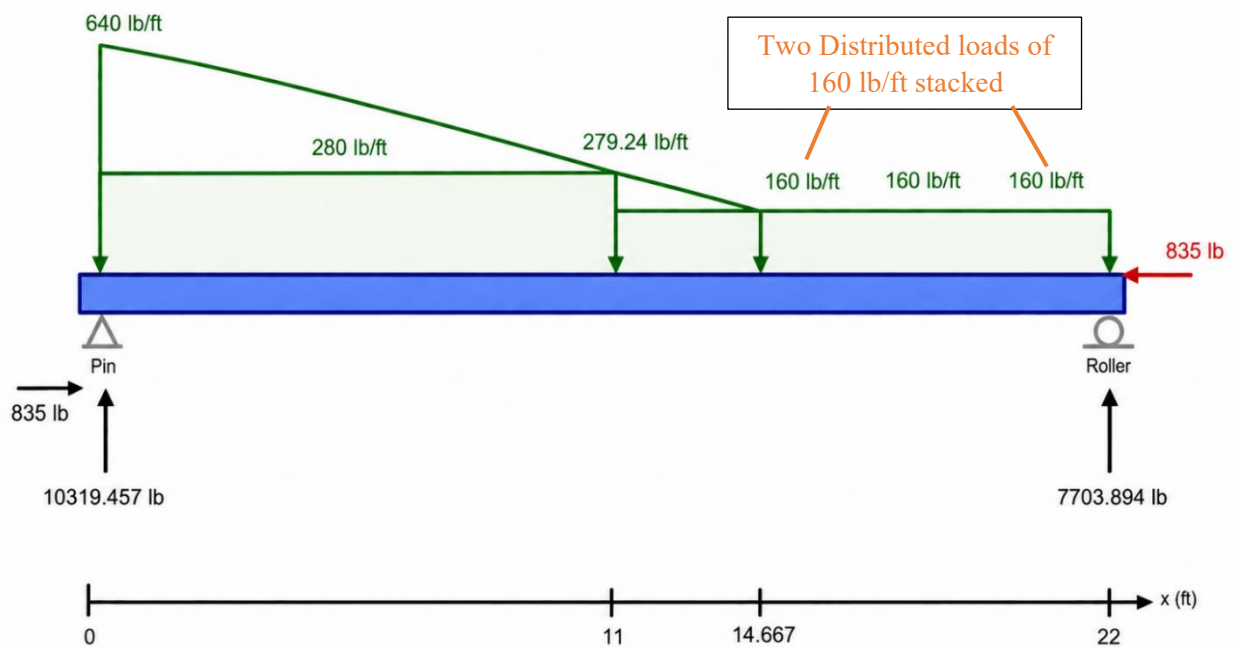


Figure 10 - Load Combination 3C: $1.2D + 1.6S + 0.8W_2$

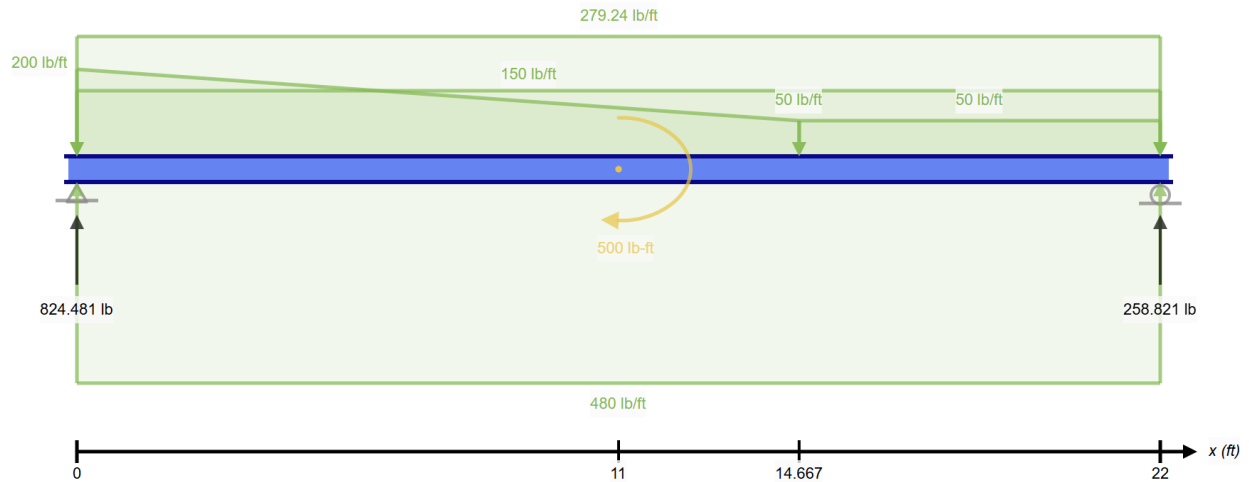


Figure 11 - Load Combination 4A: $1.2D + 1.6S + 0.8W_1$

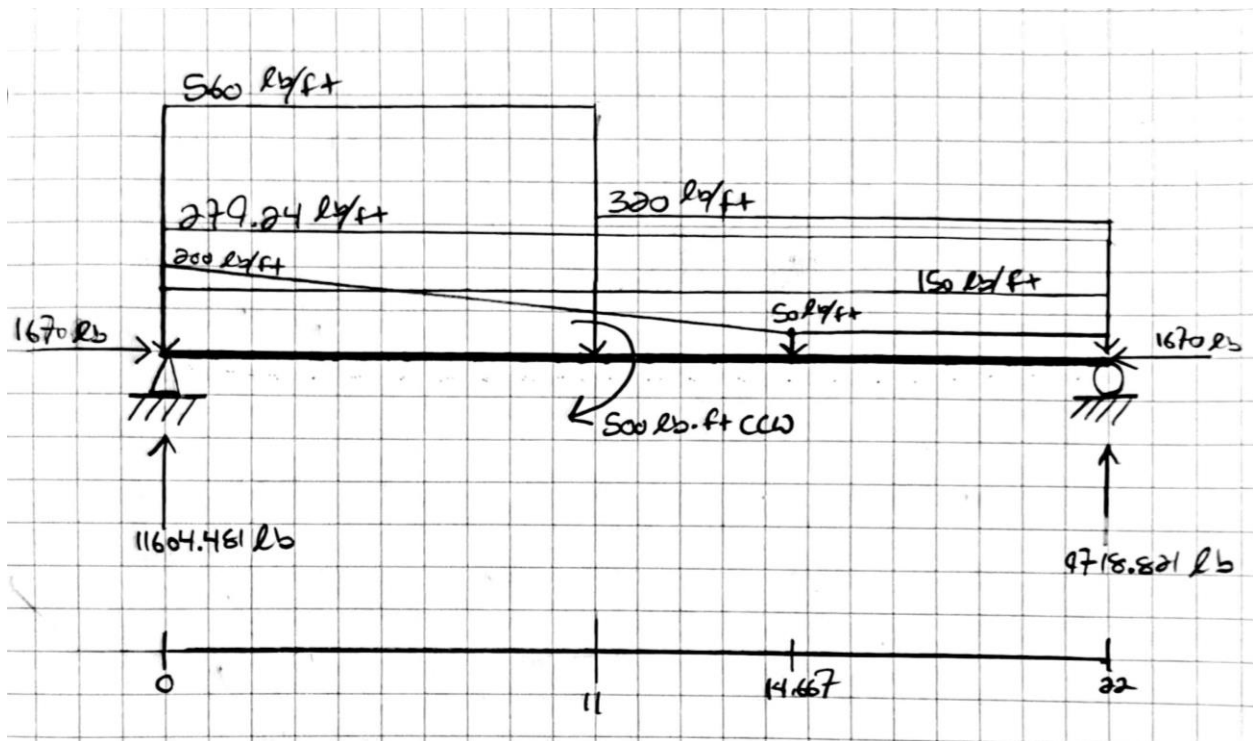


Figure 12 - Load Combination 4B: $1.2D + 1.6S + 0.8W_2$

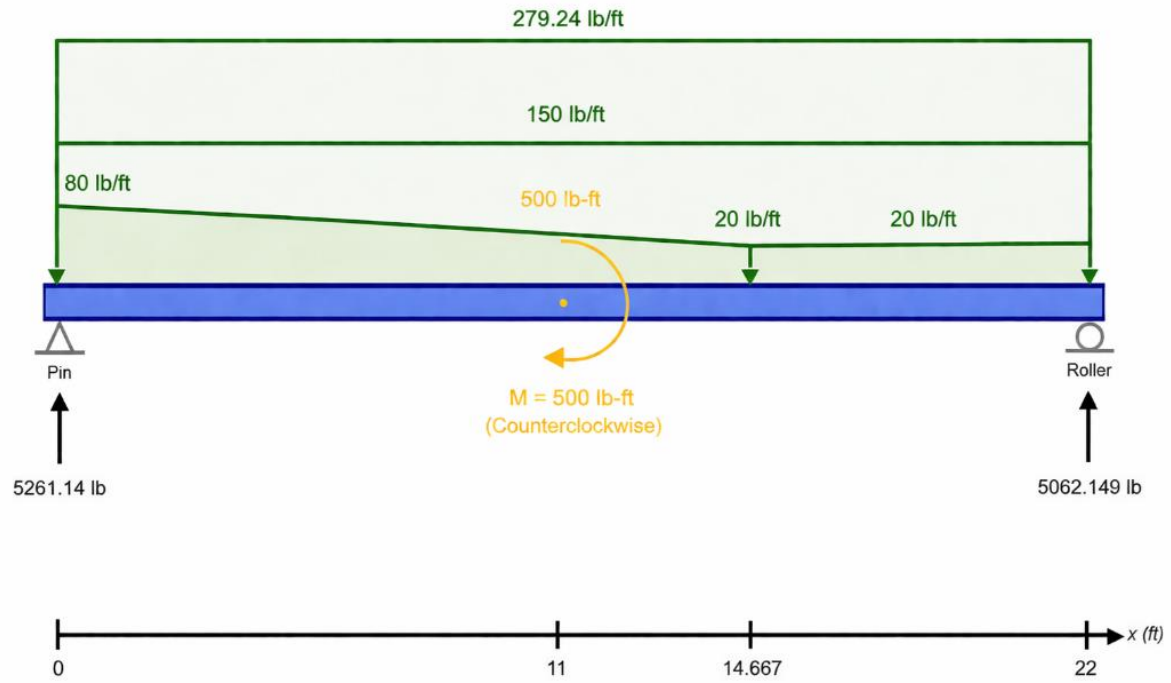


Figure 13 - Load Combination Five: $1.2D + L + 0.2S$

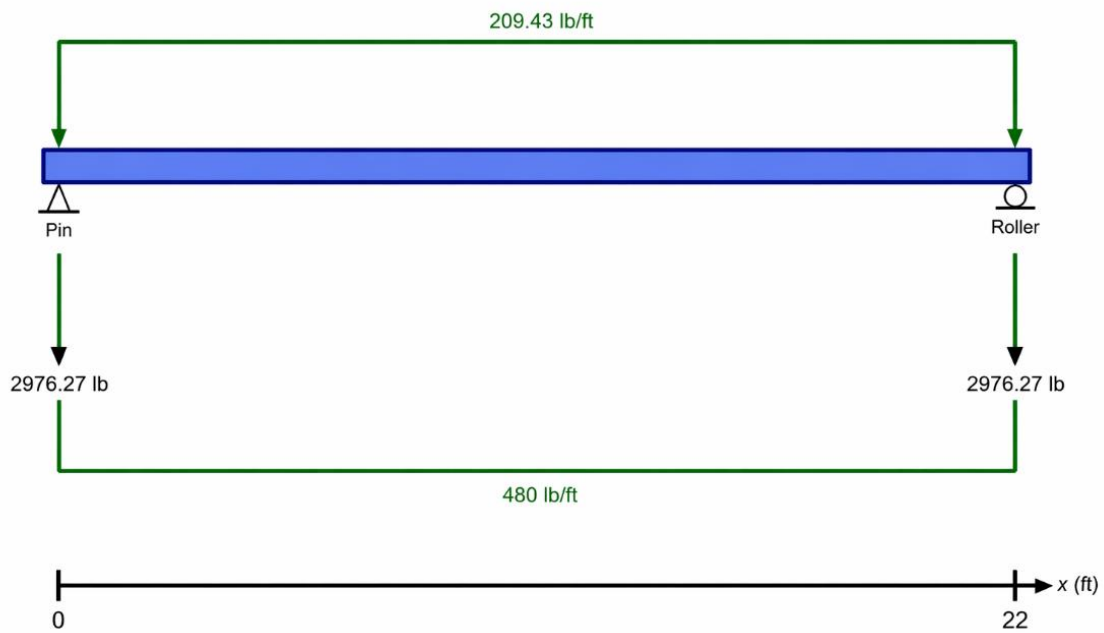


Figure 14 - Load Combination 6A: $0.9D + 1.6W_1$

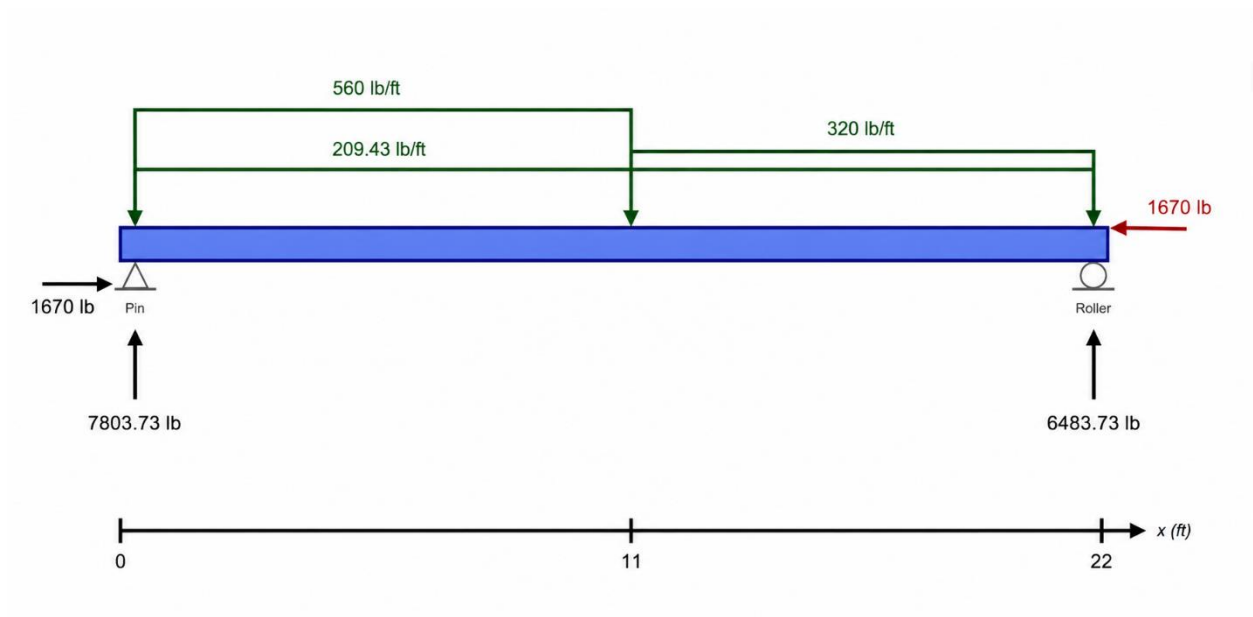


Figure 15 - Load Combination 6B: $0.9D + 1.6W_2$

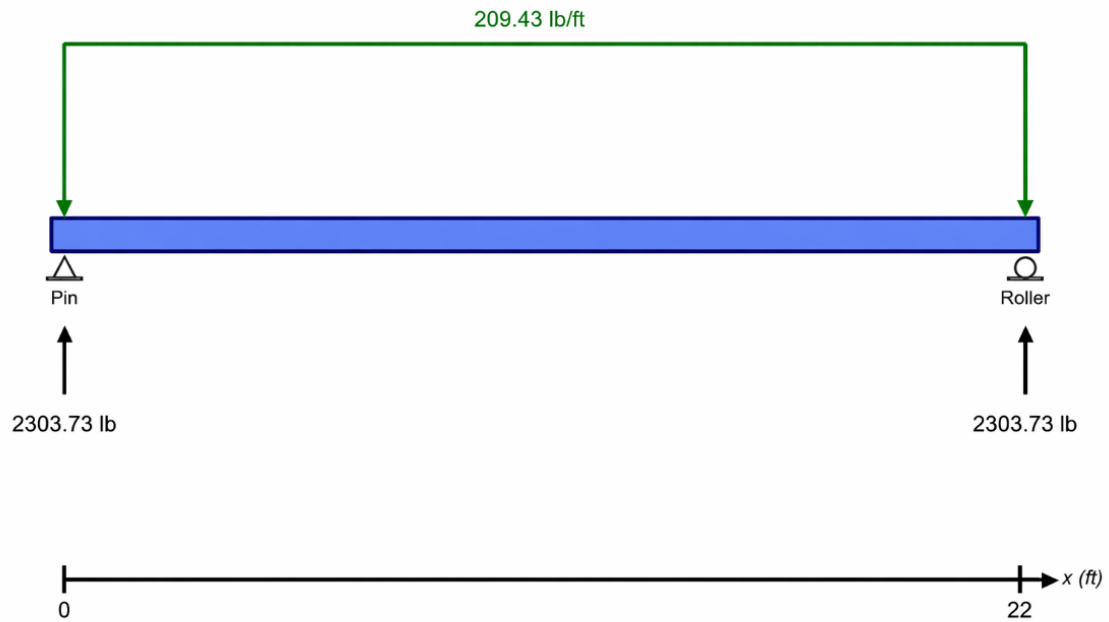


Figure 16 - Load Combination Seven: $0.9D$

Step 3: Moment and Shear Diagrams (20 pts)

(a) *Shear and Moment Diagrams*

All graph generations are done in SkyCiv:

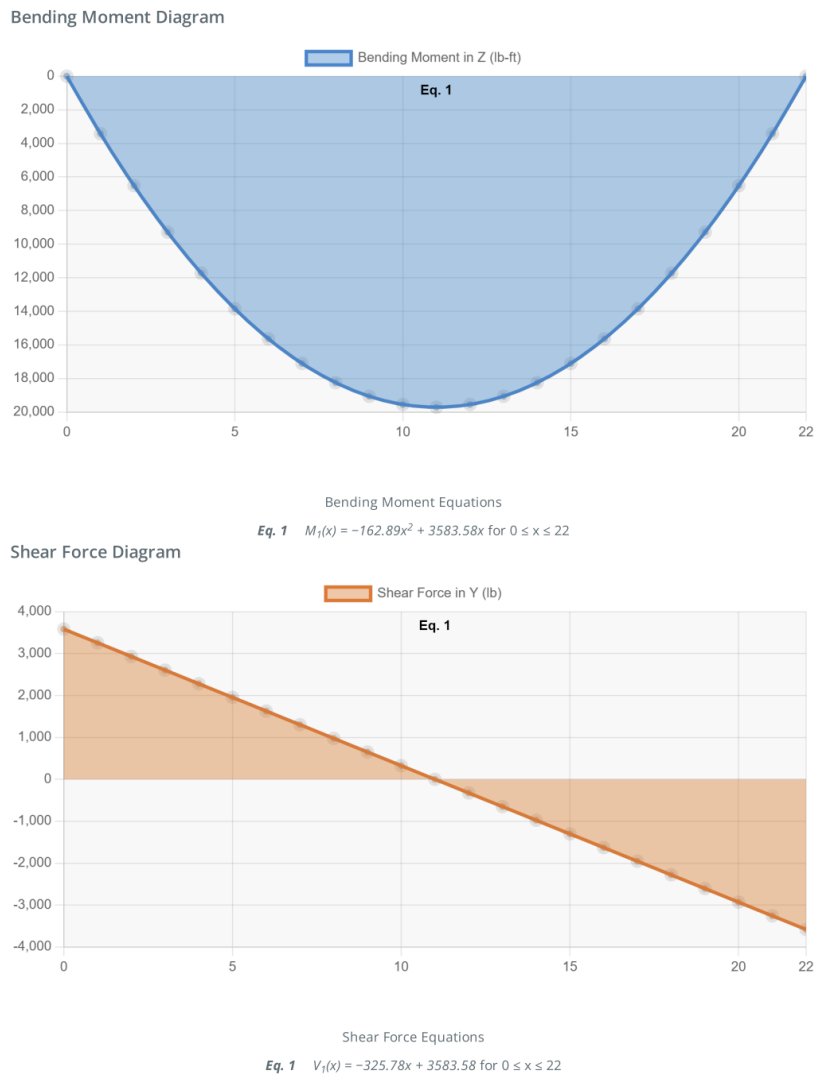
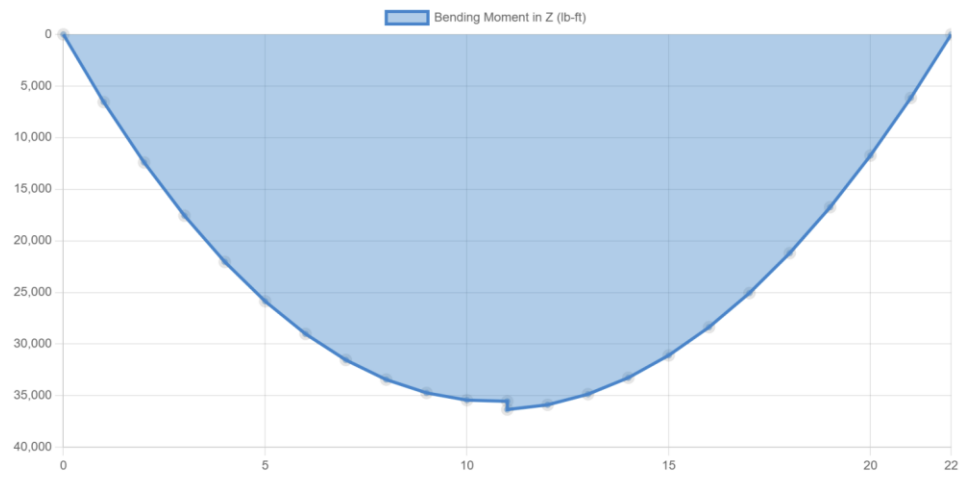


Figure 17 - Load Combination One: 1.4D

Bending Moment Diagram



Shear Force Diagram

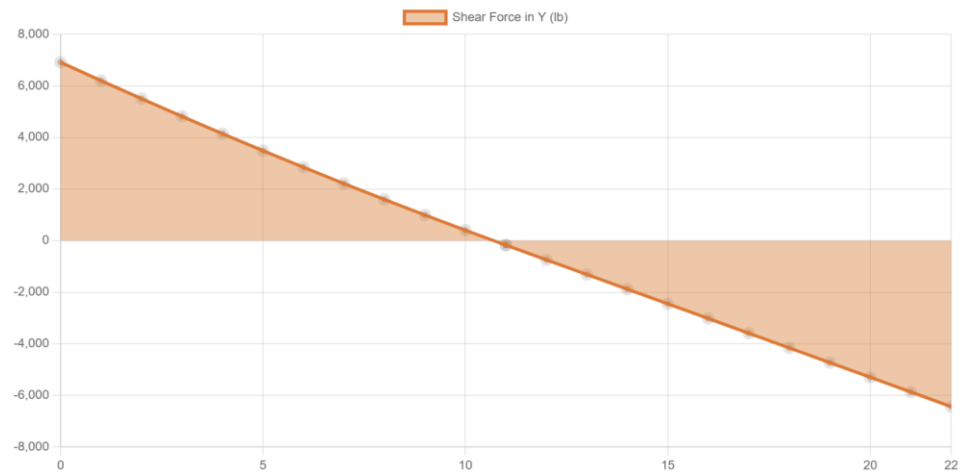
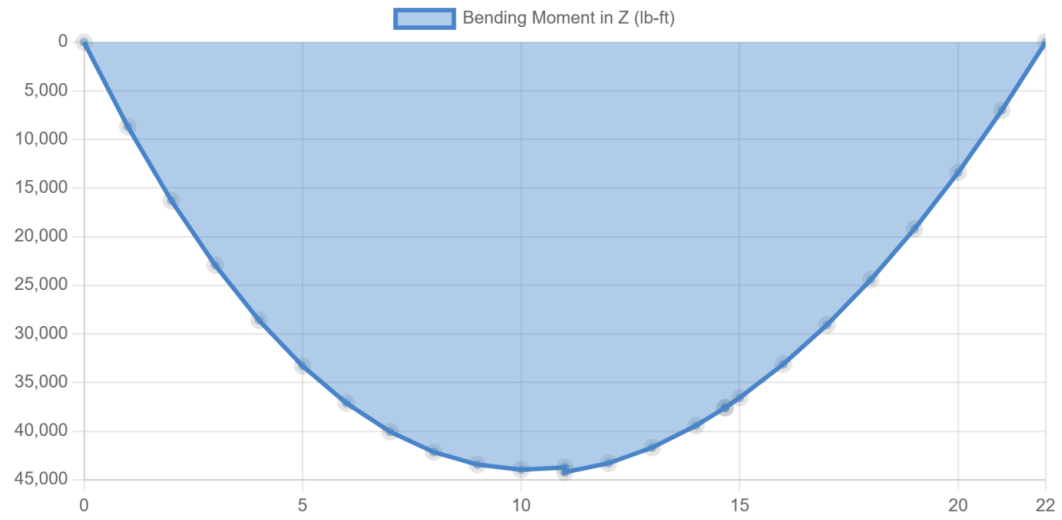


Figure 18 - Load Combination Two: $1.2D + 1.6L + 0.5S$

Bending Moment Diagram



Shear Force Diagram

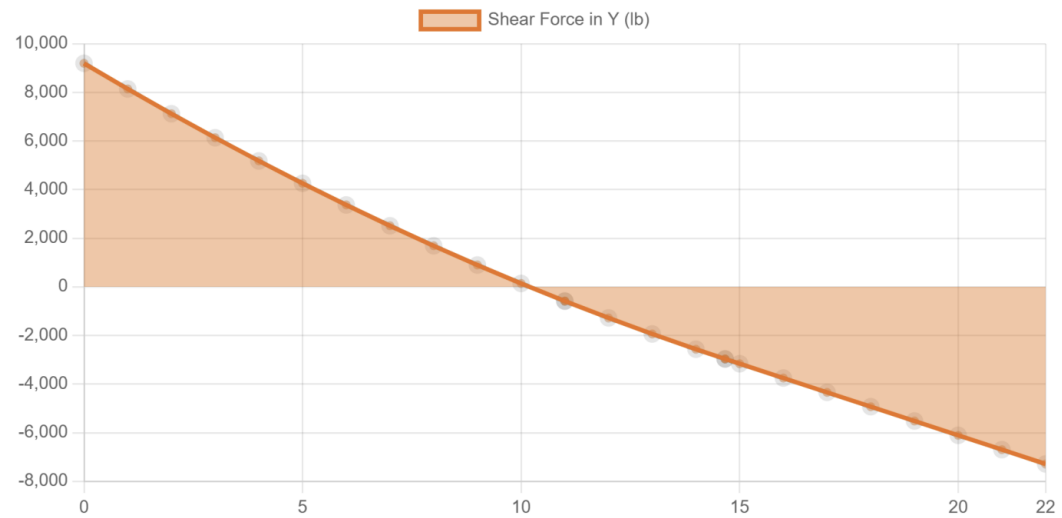
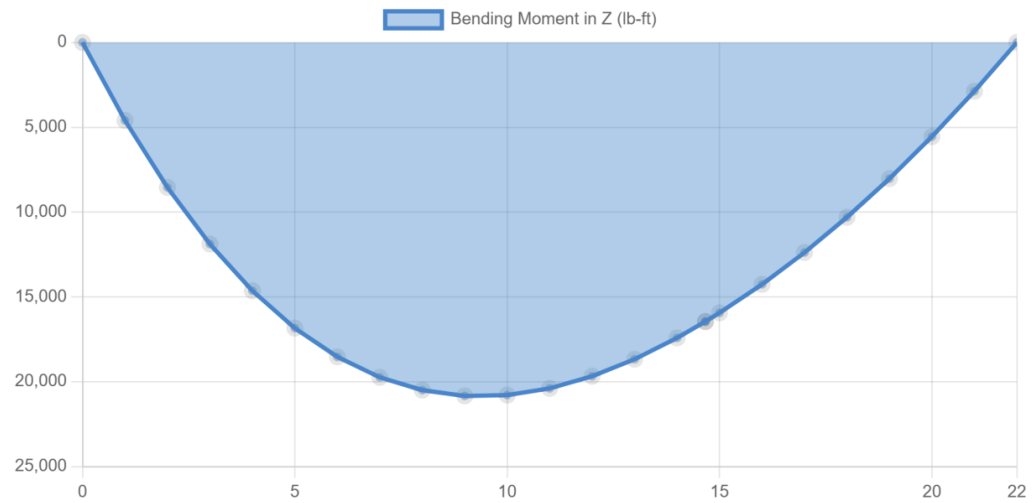


Figure 19 - Load Combination 3A: $1.2D + 1.6S + L$

Bending Moment Diagram



Shear Force Diagram

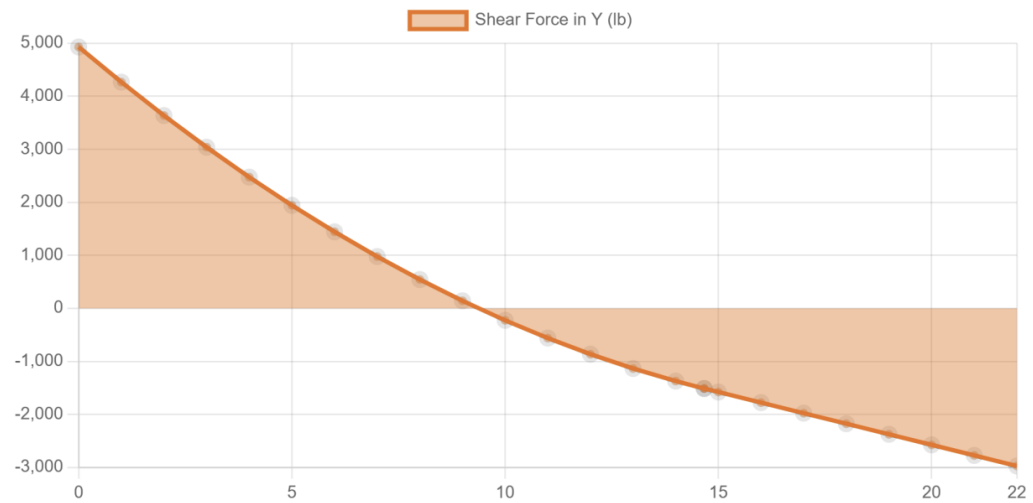
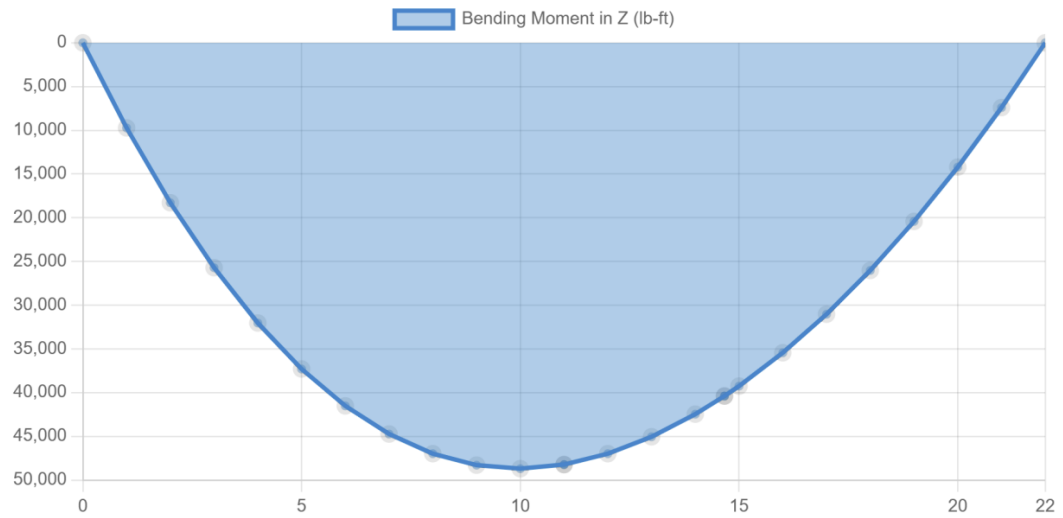


Figure 20 - Load Combination 3B: $1.2D + 1.6S + 0.8W_1$

Bending Moment Diagram



Shear Force Diagram

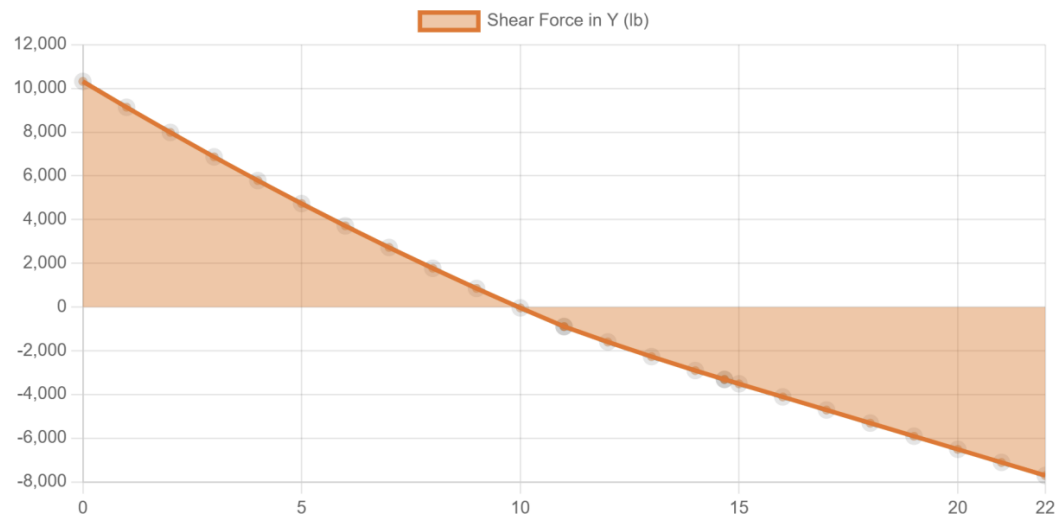
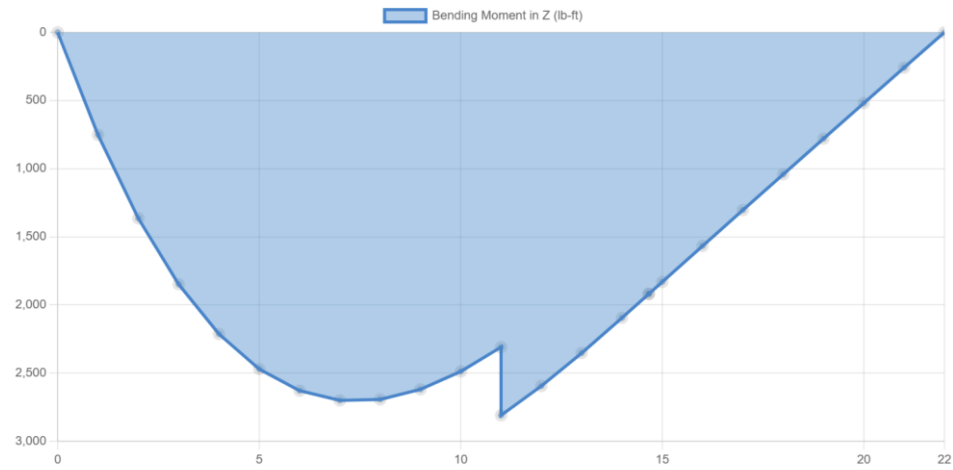


Figure 21 - Load Combination 3C: $1.2D + 1.6S + 0.8W_2$

Bending Moment Diagram



Shear Force Diagram

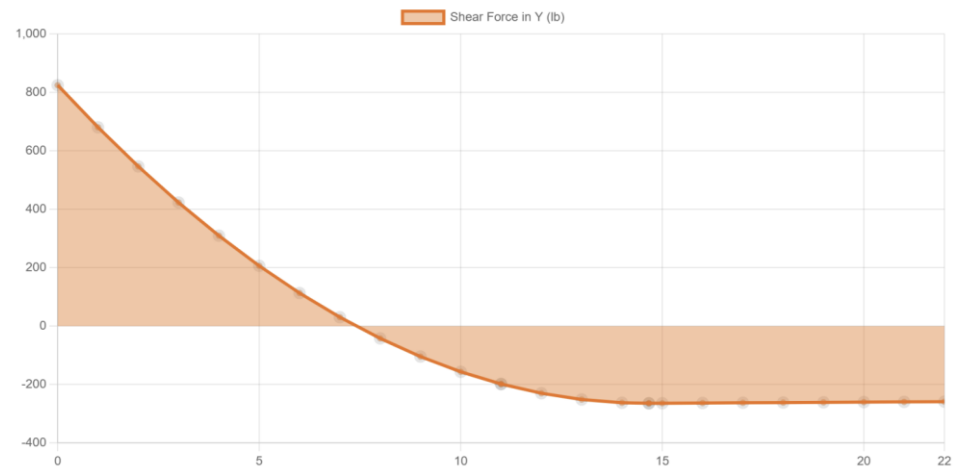
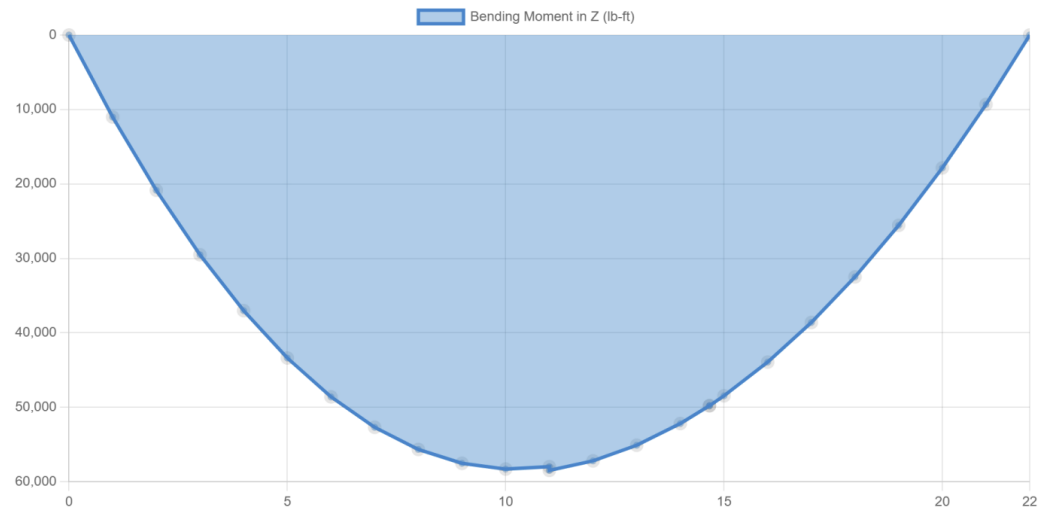


Figure 22 - Load Combination 4A: $1.2D + 1.6S + 0.8W_1$

Bending Moment Diagram



Shear Force Diagram

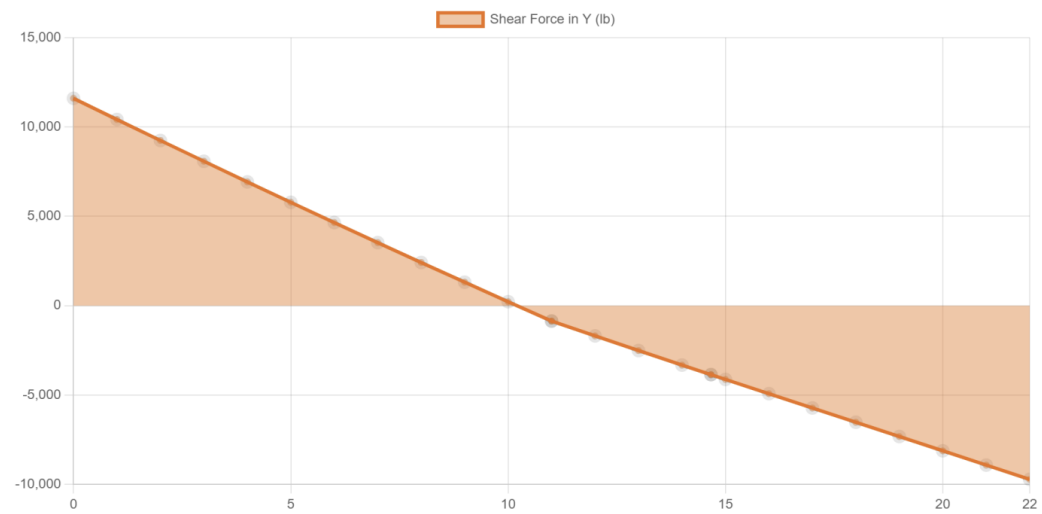
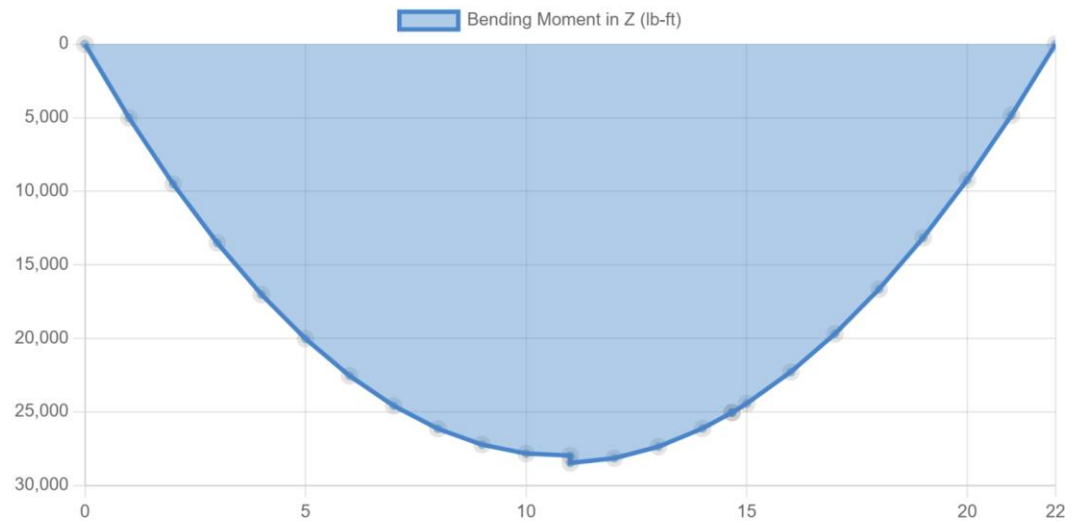


Figure 23 - Figure 12 - Load Combination 4B: $1.2D + 1.6S + 0.8W_2$

Bending Moment Diagram

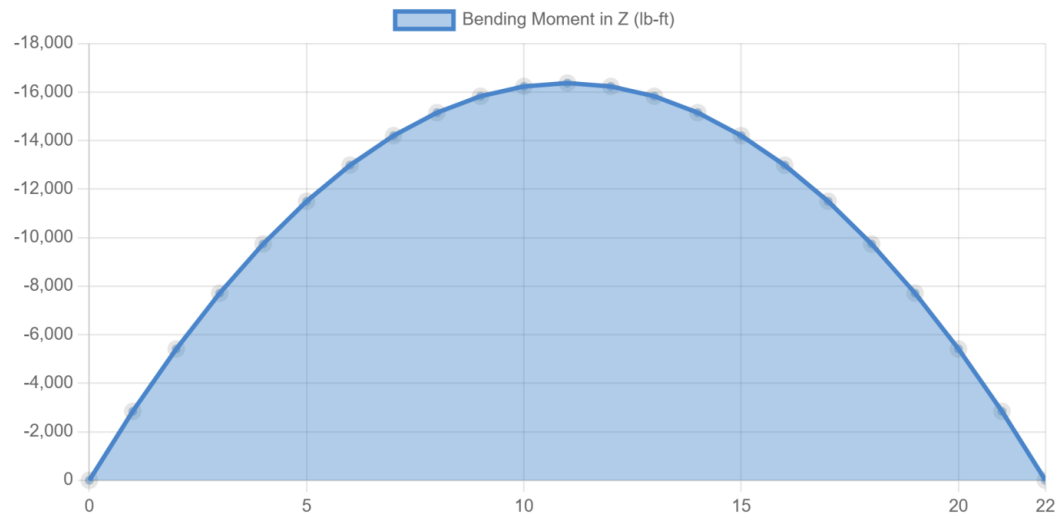


Shear Force Diagram



Figure 24 - Load Combination Five: $1.2D + L + 0.2S$

Bending Moment Diagram



Shear Force Diagram

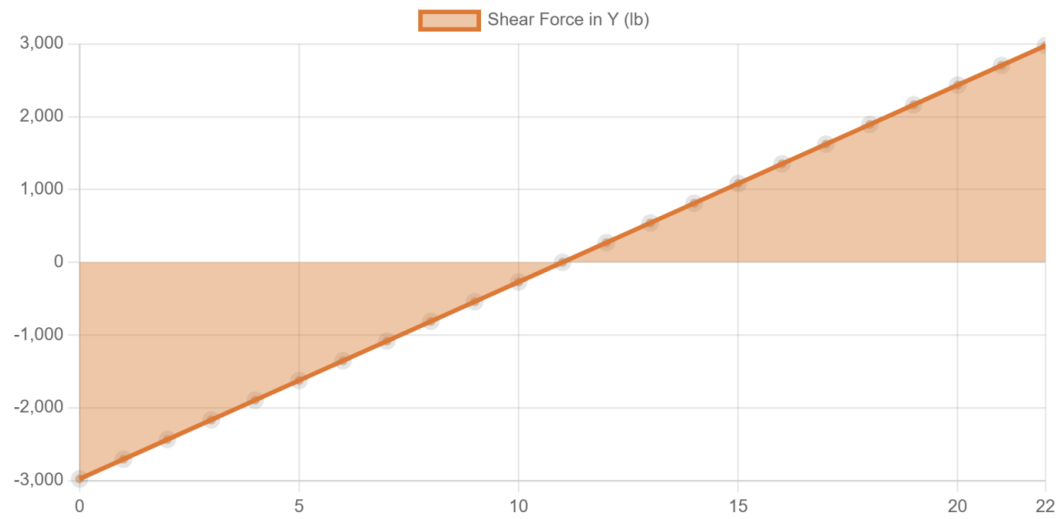
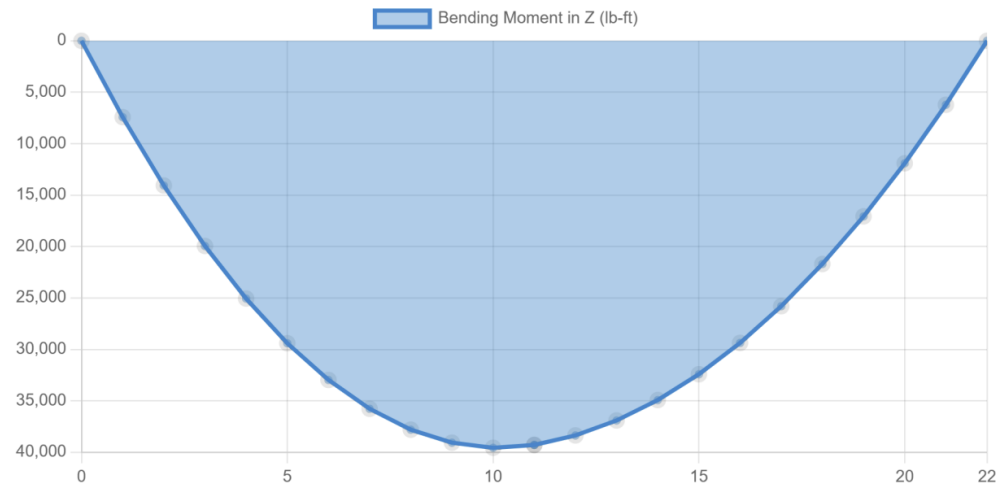


Figure 25 - Load Combination 6A: $0.9D + 1.6W_1$

Bending Moment Diagram



Shear Force Diagram

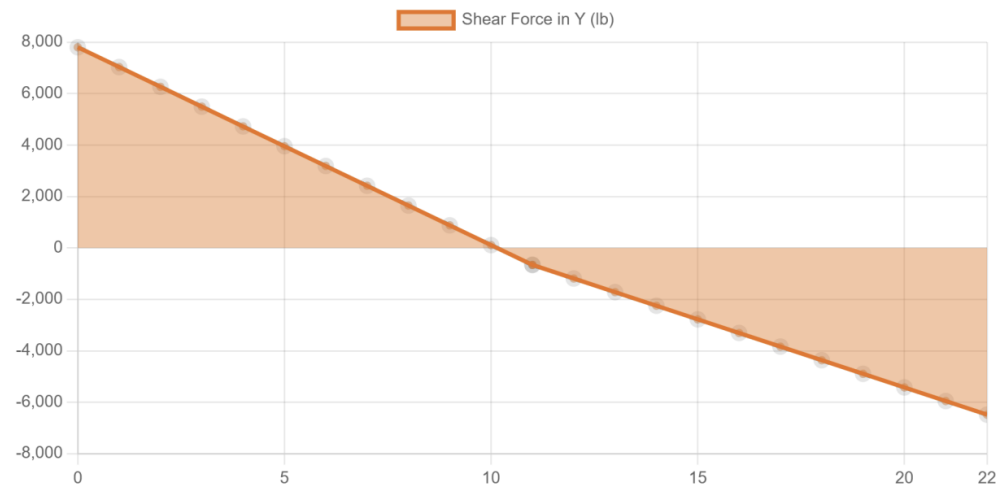
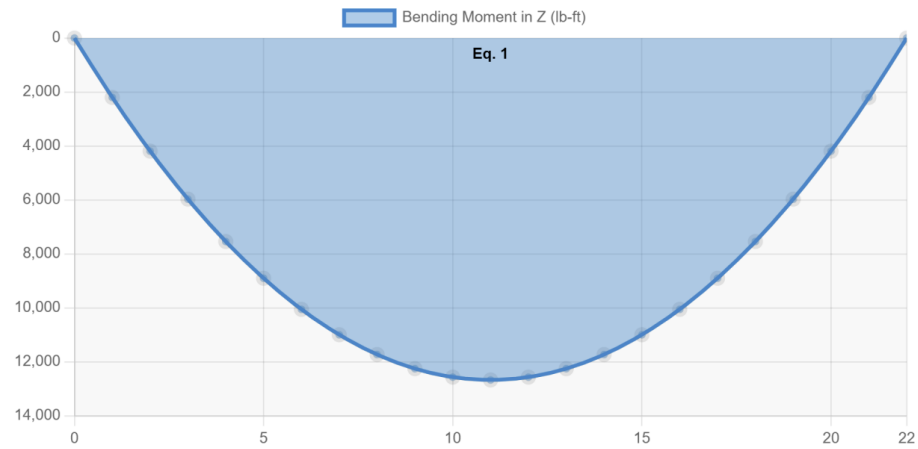


Figure 26 - Load Combination 6B: $0.9D + 1.6W_1$

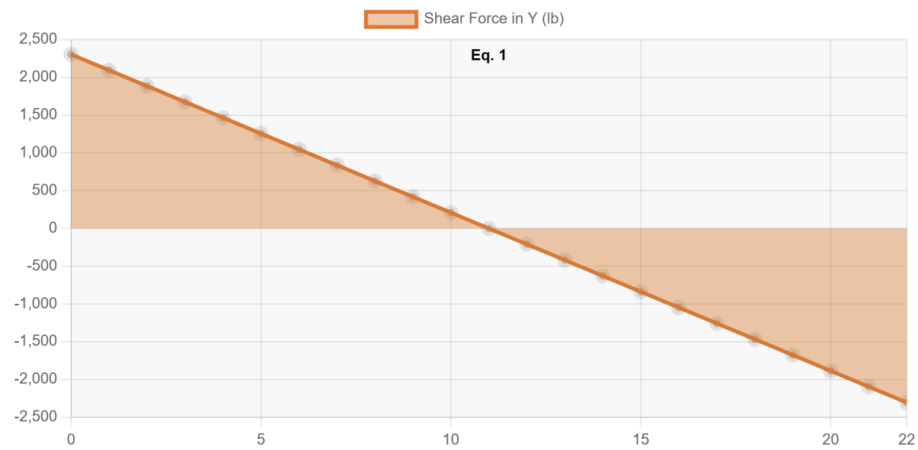
Bending Moment Diagram



Bending Moment Equations

Eq. 1 $M_1(x) = -104.715x^2 + 2303.73x$ for $0 \leq x \leq 22$

Shear Force Diagram



Shear Force Equations

Eq. 1 $V_1(x) = -209.43x + 2303.73$ for $0 \leq x \leq 22$

Figure 27 - Load Combination Seven: 0.9D

(b) *Verification of Shear and Moment Diagrams*

Load Case Seven:

Work, outside of superposition, generated in SkyCiv:

Shear for $0 \leq x \leq 22$

Take a cut for:

$$0 \leq x \leq 22$$

Find the forces due to the dead load from $x = 0$ to $x = 22$.

The rectangular dead load acts at a distance of:

$$\frac{x}{2}$$

from the cut, with a force of:

$$base \times height = -209.43x$$

Using vertical force equilibrium:

$$+ \uparrow \sum F_y = 0$$

$$+2303.73 + [-209.43x] - V_1(x) = 0$$

$$V_1(x) = 2303.73 - 209.43x$$

Therefore:

$$\boxed{V_1(x) = 2303.73 - 209.43x} \quad 0 \leq x \leq 22$$

Moment for $0 \leq x \leq 22$

Take a cut for:

$$0 \leq x \leq 22$$

Find the moments due to the dead load from $x = 0$ to $x = 22$.

The rectangular dead load acts at a distance of:

$$\frac{x}{2}$$

from the cut, with a force of:

$$base \times height = -209.43x$$

So the moment from the distributed load is:

$$(-209.43x) \left(\frac{x}{2} \right) = -104.715x^2$$

Using moment equilibrium about the cut:

$$+ \circlearrowleft \sum M_x = 0$$

$$+(2303.73)(x - 0) + [-104.715x^2] - M_1(x) = 0$$

$$M_1(x) = 2303.73x - 104.715x^2$$

Therefore:

$$\boxed{M_1(x) = 2303.73x - 104.715x^2} \quad 0 \leq x \leq 22$$

Load Case 3B:

Combined Shear and Moment Calculations

Using superposition:

$$V_{\text{total}} = V_D + V_S + V_W$$

$$M_{\text{total}} = M_D + M_S + M_W$$

For the load combination:

$$1.2D + 1.6S + 0.8W_1$$

1. Combined Shear Equations

Region 1: $0 \leq x \leq 14.667$

$$V_D = 3071.64 - 279.24x$$

$$V_S = 4497.822 - 640x + 16.363x^2$$

$$V_W = -2640 + 240x$$

Add the shear equations:

$$V_1(x) = (3071.64 - 279.24x) + (4497.822 - 640x + 16.363x^2) + (-2640 + 240x)$$

$$V_1(x) = 4929.462 - 679.24x + 16.363x^2$$

for:

$$0 \leq x \leq 14.667$$

Region 2: $14.667 \leq x \leq 22$

$$V_D = 3071.64 - 279.24x$$

$$V_S = 977.742 - 160x$$

$$V_W = -2640 + 240x$$

Add the shear equations:

$$V_2(x) = (3071.64 - 279.24x) + (977.742 - 160x) + (-2640 + 240x)$$

$$V_2(x) = 1409.382 - 199.24x$$

for:

$$14.667 \leq x \leq 22$$

2. Combined Moment Equations

Region 1: $0 \leq x \leq 14.667$

$$M_D = 3071.64x - 139.62x^2$$

$$M_S = 4497.822x - 320x^2 + 5.454x^3$$

$$M_W = -2640x + 120x^2$$

Add the moment equations:

$$M_1(x) = (3071.64x - 139.62x^2) + (4497.822x - 320x^2 + 5.454x^3) + (-2640x + 120x^2)$$

$$M_1(x) = 4929.462x - 339.62x^2 + 5.454x^3$$

for:

$$0 \leq x \leq 14.667$$

Region 2: $14.667 \leq x \leq 22$

$$M_D = 3071.64x - 139.62x^2$$

$$M_S = 17209.671 + 977.742x - 80x^2$$

$$M_W = -2640x + 120x^2$$

Add the moment equations:

$$M_2(x) = (3071.64x - 139.62x^2) + (17209.671 + 977.742x - 80x^2) + (-2640x + 120x^2)$$

$$M_2(x) = 17209.671 + 1409.382x - 99.62x^2$$

for:

$$14.667 \leq x \leq 22$$

3. Final Combined Equations

$$V(x) = \begin{cases} 4929.462 - 679.24x + 16.363x^2, & 0 \leq x \leq 14.667 \\ 1409.382 - 199.24x, & 14.667 \leq x \leq 22 \end{cases}$$

$$M(x) = \begin{cases} 4929.462x - 339.62x^2 + 5.454x^3, & 0 \leq x \leq 14.667 \\ 17209.671 + 1409.382x - 99.62x^2, & 14.667 \leq x \leq 22 \end{cases}$$

Full Calculations for each load type:

[Load 3B: Snow Load Shear and Moment Calculations](#)

[Load 3B: Dead Load Shear and Moment Calculations](#)

[Load 3B: Wind Load 1 Shear and Moment Calculations](#)

(c) Maximum and Minimum Values of Shear and Moment

Table 2 - Max and Min Values of Shear and Moment Diagrams, Shear in lb, Moment in lb·ft

Load Case	V(x) Max	V(x) Min	V(x) Zeros	M(x) Max	M(x) Min	M(x) Zeros
1	3584 x=0	-3584 x=22	x=11	19710 x=11	0 x=0, 22	x=0, 22
2	6913 x=0	-6436 x=22	x=10.7	37294 x=11	0 x=0, 22	x=0, 22
3	9197 x=0	-7287 x=22	x=10.2	44235 x=11	0 x=0, 22	x=0, 22
3B	4929 x=0	-2974 x=22	x=8.9	20832 x=10	0 x=0, 22	x=0, 22
3C	10319 x=0	-7704 x=22	x=10	48687 x=10	0 x=0, 22	x=0, 22
4A	824 x=0	-259 x=22	x=7.4	2809 x=11	0 x=0, 22	x=0, 22
<u>4B</u>	<u>11604</u> x=0	-9719 x=22	x=10.2	<u>58469</u> x=11	0 x=0, 22	x=0, 22
5	5261 x=0	-5062 x=22	x=10.8	28471 x=11	0 x=0, 22	x=0, 22
6A	2976 x=0	-2976 x=22	x=11	0 x=0, 22	-16369 x=11	x=0, 22
6B	7804 x=0	-6484 x=22	x=10.1	39566 x=10	0 x=0, 22	x=0, 22
7	2304 x=0	-2304 x=22	x=11	12671 x=11	0 x=0, 22	x=0, 22

Step 4: Stress and Strain Calculations (15 pts)

(a) Locations of the Maximum Internal Forces

Table 3 – Locations and Values of Max Internal Forces

Load Case	V(x) Max	V(x) Min	V(x) Zeros	M(x) Max	M(x) Min	M(x) Zeros
1	3584 x=0	-3584 x=22	x=11	19710 x=11	0 x=0, 22	x=0, 22
2	6913 x=0	-6436 x=22	x=10.7	37294 x=11	0 x=0, 22	x=0, 22
3	9197 x=0	-7287 x=22	x=10.2	44235 x=11	0 x=0, 22	x=0, 22
3B	4929 x=0	-2974 x=22	x=8.9	20832 x=10	0 x=0, 22	x=0, 22
3C	10319 x=0	-7704 x=22	x=10	48687 x=10	0 x=0, 22	x=0, 22
4A	824 x=0	-259 x=22	x=7.4	2809 x=11	0 x=0, 22	x=0, 22
<u>4B</u>	<u>11604</u> x=0	-9719 x=22	x=10.2	<u>58469</u> x=11	0 x=0, 22	x=0, 22
5	5261 x=0	-5062 x=22	x=10.8	28471 x=11	0 x=0, 22	x=0, 22
6A	2976 x=0	-2976 x=22	x=11	0 x=0, 22	-16369 x=11	x=0, 22
6B	7804 x=0	-6484 x=22	x=10.1	39566 x=10	0 x=0, 22	x=0, 22
7	2304 x=0	-2304 x=22	x=11	12671 x=11	0 x=0, 22	x=0, 22

To calculate the maximum normal and shear stress resulting from the maximum internal forces, our group had to identify the sources of stress, which included normal stress due to bending moment, and shear stress due to transverse shear. The formulas for finding these values for each load case are listed below as well as the values that remained the same for each of the loads, which included I_x and c for bending moment, and Q , I_x , and t , for shear stress. These values were calculated below and then multiplied by each load case's respective max moment or max shear. They were then compared with yielding stress for steel to determine whether the case would fail due to yielding, which they did not. Additionally, the Mohr's circle for the cases are drawn below as well.

(b) *Maximum Normal and Shear Stresses*

$$A_{beam} = 9.60 \text{ in}^2$$

$$\text{Bottom Flange Centroid: } y_b = 0.375 \text{ in}$$

$$\text{Web Centroid: } y_w = 0.75 + \frac{9.5}{2} = 5.50 \text{ in}$$

$$\text{Top Flange Centroid: } y_t = 11 - 0.375 = 10.625 \text{ in}$$

$$\bar{y} = \frac{(2.25 \text{ in}^2)(0.375 \text{ in}) + (2.85 \text{ in}^2)(5.50 \text{ in}) + (4.50 \text{ in}^2)(10.625 \text{ in})}{9.60 \text{ in}^2}$$

$$\bar{y} = 6.70 \text{ in above the bottom}$$

$$I = I_c + Ad^2$$

Bottom Flange:

$$I_{b,c} = \frac{3(0.75)^3}{12} = 0.1055 \text{ in}^4$$

$$d_b = 6.701 - 0.375 = 6.326 \text{ in}$$

$$I_b = 0.1055 + 2.25(6.326)^2 = 90.13 \text{ in}^4$$

Web:

$$I_{w,c} = \frac{0.30(9.50)^3}{12} = 21.434 \text{ in}^4$$

$$d_b = 6.701 - 5.50 = 1.201 \text{ in}$$

$$I_b = 21.434 + 2.85(1.201)^2 = 25.54 \text{ in}^4$$

Top Flange:

$$I_{t,c} = \frac{6(0.75)^3}{12} = 0.2109 \text{ in}^4$$

$$d_b = 10.625 - 6.701 = 3.924 \text{ in}$$

$$I_b = 0.2109 + 4.50(3.924)^2 = 69.52 \text{ in}^4$$

Total:

$$I_x = 90.13 + 25.54 + 69.52 = 185.15 \text{ in}^4$$

Example Calculations:

$$\sigma_{max} = \frac{M_{max}c}{I_x} = \frac{M_{max}(lb \cdot ft) \left(12 \frac{in.}{ft}\right) (6.7in.)}{185.15 \text{ in}^4}$$

Yielding stresses:

$$\sigma_{yielding, steel} = 36 \cdot 10^3 \frac{lb}{in^2}$$

$$\tau_{yielding, steel} = .5\sigma_{yielding} = 18 \cdot 10^3 \frac{lb}{in^2}$$

Shear Stress calculations:

$$\begin{aligned}
 Q_t &= A \cdot y = (6in \cdot 0.75in) \left(\left(11in. - \frac{.75}{2}in. \right) - 6.7in \right) = 4.5in^2(3.925in.) \\
 &= 17.66in^3 \\
 Q_w &= A \cdot y = (.3in \cdot 3.55in) \left(\frac{3.55}{2}in. \right) = 1.89in^3 \\
 Q_1 &= Q_t + Q_w = 17.66in^3 + 1.89in^3 = 19.55in.^3 \\
 \tau_{max} &= \frac{V_{max}Q}{I_x t} = \frac{V_{max}(19.55in.^3)}{(185.15in.^4)(0.3in)}
 \end{aligned}$$

Table 4 - Max Shear, Moment, and Accompanying Normal and Shear Stress

Load Case	V Max	M Max	Normal Stress Max(psi)	Shear Stress Max(psi)
1	3584 x=0	19710 x=11	8559	1261
2	7094 x=0	37294 x=11	16194	2497
3	9197 x=0	44235 x=11	19209	3237
3B	4929 x=0	20832 x=10	9046	1735
3C	10319 x=0	48687 x=10	21142 + 87	3632
4A	824 x=0	2809 x=11	1220	290
4B	11604 x=0	58469 x=11	25390 + 174	4084
5	5261 x=0	28471 x=11	12363	1852
6A	2976 x=0	-16369 x=11	7108	1047
6B	7804 x=0	39566 x=10	17181+ 174	2747
7	2304 x=0	12671 x=11	5502	811

(c) Principal Stresses and Absolute Maximum Shear Stress, Failure under Normal and Shear Stress

To calculate the following absolute shear stress and principal stresses, our group used the principal stress equations to obtain our new values, which are recorded on the following table:

Table 5 - Principal Stresses

Load Case	σ_1	σ_2	τ_{abs}
1	-182	8741	4461
2	-376	16570	8473
3	-531	19739	10135
3B	-321	9367	4844
3C	-606	21748	11177
4A	-65.4	1285	675
4B	-640	26030	13335
5	-271	12634	6453
6A	-151	7259	3705
6B	-429	17610	9019
7	-114	5747	2930

For all cases, normal stress does not exceed the yielding stress of 36000 psi, indicating the beam will not fail under normal stress.

*For all cases, shear stress does not exceed 18000psi, so it will not fail under shear stress either***

Our group then drew the elements of the principal stresses for two of the load combinations on a graph to get Mohr's circle for these two cases. In this we calculate the average normal stress to obtain the center and plotted our normal and shear stress to get our max shear stress and max principal stresses. For Case 4b, we obtained a radius equal to 13335 psi, which is also the same value of the max shear stress.

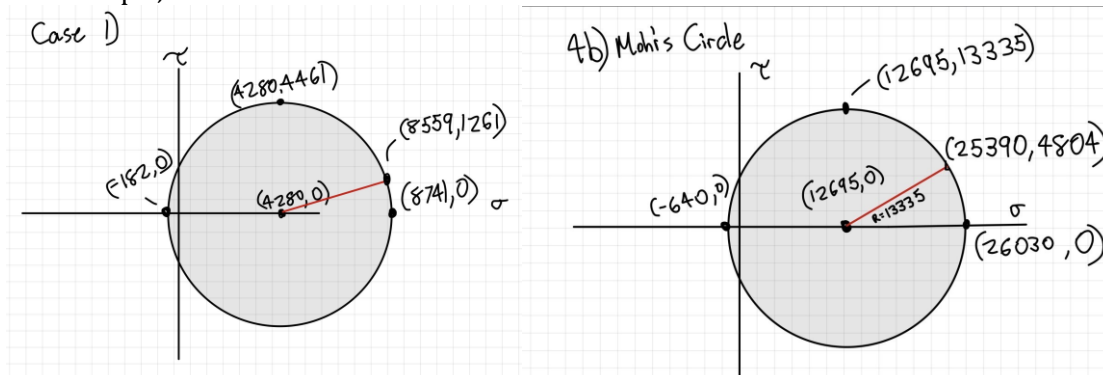


Figure 28 - Load Combination 1 & 4b Mohr's Circle showing maximum stress location

(d) Three Steel Plates

Three steel plates welded together would need to sustain at least $3690 \frac{lb}{in^2}$. The following calculations show the stress calculated by welding the plates on the bottom and on the top flange with the web, by shearing the top flange and shearing the bottom flange.

$$\tau = \frac{VQ}{It} = \frac{V}{It} A_{top} \cdot y_{top} = \frac{11604lb \cdot (17.66in^3)}{185.15in^4 \cdot 0.3in} = 3690 \frac{lb}{in^2}$$

$$\tau = \frac{VQ}{It} = \frac{V}{It} A_{bottom} \cdot y_{bottom} = \frac{11604lb \cdot (14.23in^3)}{185.15in^4 \cdot 0.3in} = 2972 \frac{lb}{in^2}$$

(e) Strain Rosette.

Regarding the strain rosettes, our team determined that locating a strain rosette configuration of 0-45-90 degrees at $x = 11ft$ would show us the readings when the beam undergoes a decent amount of strain. The following calculations were made to find the strain rosette readings that should be measured at $x = 11ft$ for case load 4b. It is important to note that from these calculations we would place the rosettes on the top of the outer surface, in order to have a negligible transverse shear stress. This results in $\gamma_{xy} = 0$, making it easier to predict the readings of the rosettes. From this point, the calculations for predicting the rosettes readings can be solved using Hooke's Law.

$$\epsilon_x = \frac{\sigma_x}{E} = \frac{25390 \frac{lb}{in^2}}{29 \cdot 10^6 \frac{lb}{in^2}} = 875\mu\epsilon$$

$$\epsilon_y = -\nu\epsilon_x = -0.32(875\mu\epsilon) = -280\mu\epsilon$$

Using 0-45-90 rosette at $x = 11ft$ (max moment), predicted strain readings:

$$\epsilon_0 = 875\mu\epsilon$$

$$\epsilon_{45} = \frac{(875 - 280)}{2} = 298\mu\epsilon$$

$$\epsilon_{90} = -280\mu\epsilon$$

Step 5: Deflection Calculations (15pts)

(a) *The deflection curve equation*

For the combination with the largest effect on the beam, our team agreed that the load case 4b had the largest effect with the largest Moment and shear in comparison to all the other caseloads. These were $M_{\max} = 58469 \text{ lb}\cdot\text{ft}$ at 11ft, and $V_{\max} 11604\text{lb}$.

To determine the deflection curve equation for this case load, our team utilized sky civ's interactive beam calculator to input our loads and solve for our deflections. Sky Civ uses the double integration method of the moment equation in order to determine the deflection curve, where $\frac{d^2v}{dx^2} = \frac{M(x)}{EI}$ From here we could use the superimposition principle to add them all together to obtain our total deflection curve equation. For this equation, we know that at the reactions $v(0)$ and $v(22)$ will

equal to 0 (0 deflection at reactions). Additionally, using double integration and superposition, our deflection equation was found to be:

$$v_{4b}(x) = 1.2D(x) + 1.6S(x) + 0.8W(x)$$

Using integration, we can determine that the entire deflection given for this case load will result in:

$$v(x) = \left(\frac{144}{1000EI} \right) \begin{cases} 0.13636x^5 - 45.9620x^4 + 1722.479x^3 - 381412x & 0 \leq x \leq 11 \\ 0.13636x^5 - 37.9620x^4 + 1370.48x^3 + 5808x^2 - 424004x & 11 \leq x \leq 14.667 \\ -27.96x^4 + 1077.14x^3 + 10110.42x^2 - 455556.4x + 209681 & 14.667 \leq x \leq 22 \end{cases}$$

(b) *The largest vertical deflection of the beam.*

Lastly, using this function we can also determine that the maximum deflection occurs at $x = 11$ ft, which we can calculate to be in the downward direction and is

$$v_{max} = -5.835in$$

The following graph from SkyCiv shows the maximum deflection at $x = 11$ ft, which supports our results.



Figure 29- Deflection Curve 4B

Step 6: Buckling Calculations (10pts)

The following table shows each caseload's max shear as well as the reactions of A and B. From this table we can see that the largest support reaction is from case load 4b, which is 11604 from Reaction A. We can first calculate the buckling load and from there, we can compare the value to the maximum reaction load, and if the reaction results in no buckling we can assume that none of the reactions are large enough to result in buckling, and that our column would not buckle under any of the load cases.

Additionally, the following calculations are shown for how our group calculated the buckling load. First, we assumed Pmax to be our largest reaction load, 11604lb. Then we calculated our Inertia at its weakest axis(for the 12"x8"), and used Euler's buckling equation to

solve for the rest, where we assumed $k=0.7$, and used other values that we have been using throughout this report for E and L. Our total buckling load resulted in 805,000 lb, which our support reaction does not exceed, therefore none of our support from any of our case loads lead to buckling.

$$P_{max} = 11604 \text{ lb}$$

$$I_{min} = \frac{1}{12} b \cdot h^3 = \frac{1}{12} (12 \text{ in})(8 \text{ in})^3 = 512 \text{ in}^4$$

$$P_{cr} = \frac{EI\pi^2}{(KL)^2}$$

$$P_{cr} = \frac{EI\pi^2}{(KL)^2} = \frac{(1.9 \cdot 10^6)(512 \text{ in}^4)\pi^2}{\left(0.7 \cdot 13 \text{ ft} \cdot 12 \frac{\text{in}}{\text{ft}}\right)^2} = 805 \cdot 10^3 \text{ lb}$$

$$P_{cr} = 805 \cdot 10^3 \text{ lb} > 11604 \text{ lb}$$

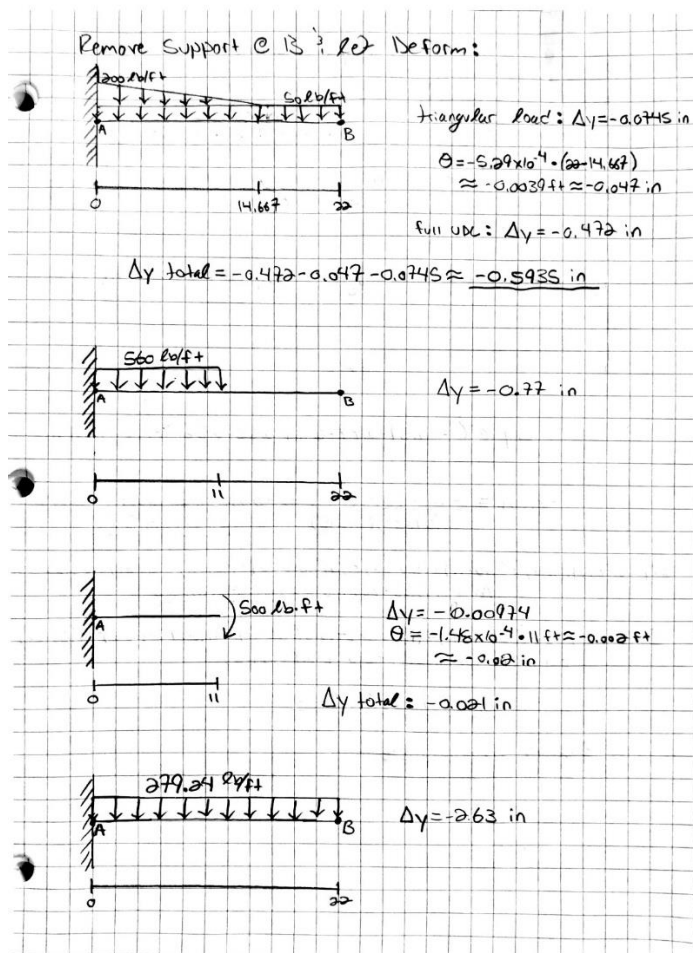
Load Case	V(x) Max	Reaction A	Reaction B
1	3584 x=0	3583.58	3583.58
2	7094 x=0	7094	6529
3	9197 x=0	9196.73	7286.62
3B	4929 x=0	4929.45	2973.89
3C	10319 x=0	10319.45	7703.89
4A	824 x=0	824.481	258.82
<u>4B</u>	<u>11604</u> x=0	<u>11604</u>	<u>9719</u>
5	5261	5261.14	5062.15

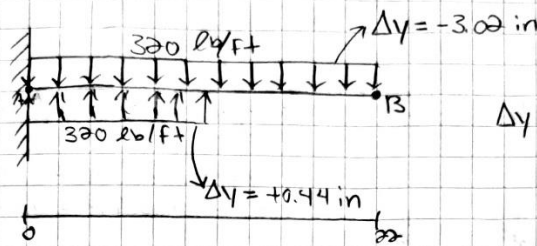
	x=0		
6A	2976	2976.27	2976.27
	x=0		
6B	7804	7803.73	6843.73
	x=0		
7	2304	2303.73	2303.73
	x=0		

Figure 30- Load Combination Max Shear and Reaction Forces

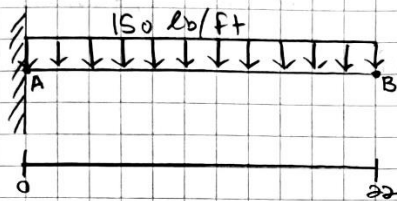
Step 7: Change in support conditions (15pts)

(a) Reactions on the Fixed and Roller Supports





$$\Delta y_{\text{total}} = -3.02 + 0.44 = -2.58 \text{ in}$$



$$\Delta y_{\text{total}} = -1.41 \text{ in}$$

$$\Delta y_{\text{all}} = -0.59 - 0.77 - 0.021 - 2.63 - 2.58 - 1.41 = -8.001 \text{ in}$$

$$-8.001 = \frac{-B_y \cdot (264)^3}{3(29 \times 10^6)(165.15)} \Rightarrow -0.00118 B_y = -8.001$$

$$B_y \approx 7004.5 \text{ lb}$$

$$\Sigma F_y = 0: 7004.5 + A_y = (320 \cdot 11) + (150 \cdot 22) + (279.24 \cdot 22) + (560 \cdot 11) + (50 \cdot 22) + \frac{1}{2}(150 \cdot 14.667)$$

$$A_y = 21330.6 - 7004.5 \Rightarrow A_y \approx 14326.1 \text{ lb}$$

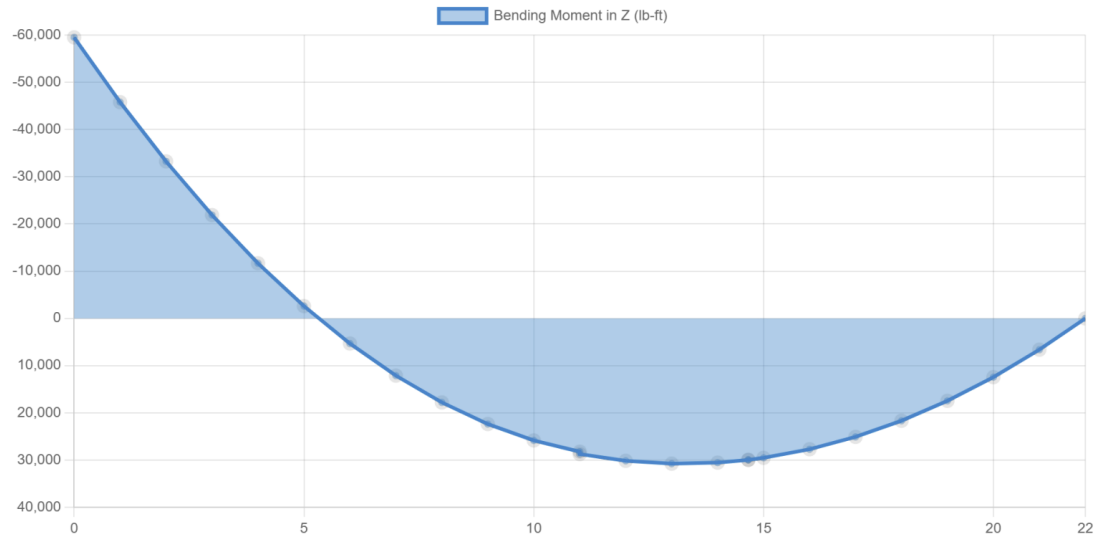
$$\Sigma M_A = 0: M_A - 5.5(560 \cdot 11) - 11(150 \cdot 22) + 16.5(320 \cdot 11) + 11(279.24 \cdot 22) - 11(50 \cdot 22) + 4.667(\frac{1}{2}(150 \cdot 14.667)) + 500 + 22(7004.5)$$

$$M_A \approx 59507.6 \text{ lb} \cdot \text{ft}$$

$$\Sigma F_x = 0: A_x = 1670 \text{ lb}$$

(b) *Shear and Moment Diagrams*

Bending Moment Diagram



Shear Force Diagram

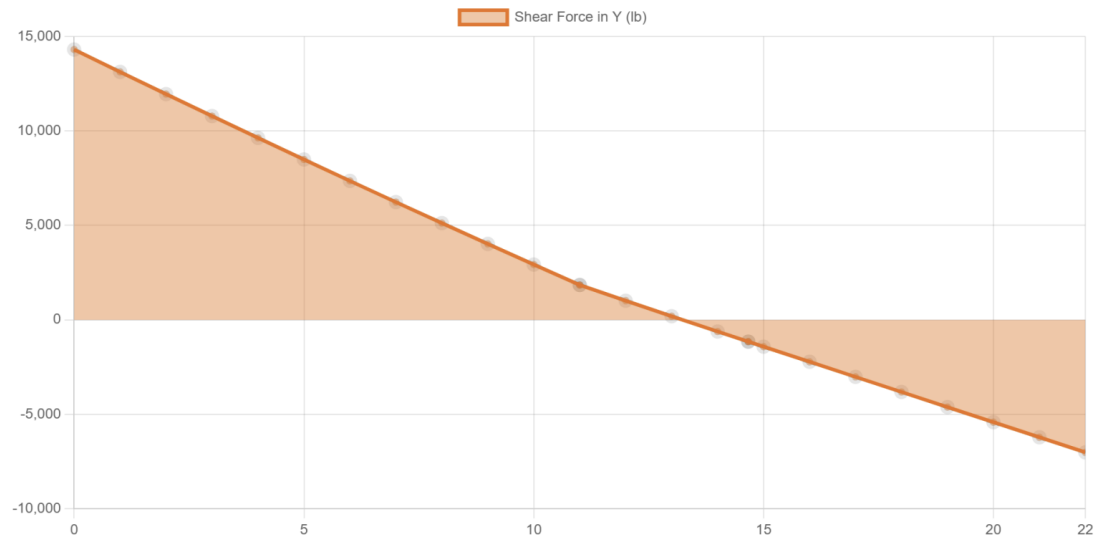


Figure 31- New Shear and Moment Diagram for 4B with Fixed Support

Changing the support of the left end to a fixed support from a pin makes the beam a propped-up cantilever instead of a simply supported beam. This creates the biggest effect on the moment diagram. With a pin, $M_A = 0$, but with a fixed support, a reaction moment at A is created with $M_A = 59,507.6 \text{ lb} \cdot \text{ft}$. This means

the moment diagram no longer starts at zero; it instead starts at $59,507.6 \text{ lb} \cdot \text{ft}$. This means because the fixed support resists rotation, there is a negative (CCW) moment area near the left column. The shear diagram changes slope according to the distributed loads because the vertical reaction forces change to $A_y = 14326.1 \text{ lb}$ and $B_y = 7004.5 \text{ lb}$. The new shear diagram holds more weight at point A, so the diagram starts at a higher point, and the zero shear point shifts closer to the left side of the beam.

(c) What Would We Advise?

We recommend the pinned-left support design. This configuration is statically determinant, meaning the reactions are straightforward to calculate and the load path clear. It avoids introducing a moment at the beam-column connection, so the left column only needs to resist axial and shear forces rather than a large bending moment. This results in simpler connection detailing, reduced construction cost, and a more forgiving system if there is settlement or thermal movement. While the beam may experience slightly larger midspan deflection and bending, this is typically manageable and more appropriate for a simple shed-type structure.

In contrast, making the left support a fixed one created a propped cantilever style beam, which is statically indeterminate and introduces a significant fixed-end moment at the column. This reduces midspan bending and deflection but shifts a large demand into the beam-column connection and into the column itself. As a result, the column and its foundation must be designed to resist combined axial load and bending, requiring more complex detailing, stronger materials, and thus higher construction costs. For this type of structure, the added stiffness is not worth the increased design complexity and demand on the frame, so the pinned support system is the more practical and efficient choice.

Conclusion

Our total analysis demonstrates how different load combinations influence the internal forces within the beam and highlights the importance of identifying critical locations for maximum shear and bending moment. Comparing support conditions showed that while fixed support increases stiffness and reduces midspan deflection, it introduces significant construction challenges designers must think about. Overall, a total structural analysis such as the one conducted in this project shows how many different problems must be addressed to come to a single solution in engineering, and how diligent we as engineers must be in order solve problems, effectively and efficiently.